Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	2 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will explain how emerging technologies can change design and manufacture. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will evaluate a technology using benefits, limitations and stakeholder needs. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	1 September 2026	By the end of this lesson, students will be able to explain how emerging technologies can change design and manufacture.	**H:** Compare an individually cut card shape with a sheet of identical computer-cut shapes. Ask what changes when production must make many consistent items rather than one model. **ID:** Use the book's new and emerging technologies section to introduce automation, computer-aided design and manufacture, and additive manufacture. Explain that a digital design is information while manufacture produces a physical object. Discuss effects on consistency, skills and the way work is organised. **WD:** Model a small manufacturer comparing manual cutting with a computer-controlled process. Trace design file, machine instructions, material and output. Identify an advantage in	**Support:** Provide a process diagram and vocabulary bank. Use one familiar school-production example before moving to the manufacturing cases, and support students in separating the machine from its digital instructions. **Extension:** Students compare suitability for a one-off prototype and a repeated production run, explaining why the same technology may not be the best choice in both situations.

			repeatability alongside setup cost and the need for trained people to maintain and check the process. **YD:** Groups study teacher-prepared cases of automated packaging and 3D-printed prototypes. For each they identify what changed, who is affected and a possible benefit or limitation. Students then independently explain one change using a specific stage of the manufacturing process. **C:** Ask students to distinguish CAD from CAM and explain why automation does not remove the need for quality checks. Collect a case response to check that the claimed effect follows from the technology described.	
2	2 September 2026	By the end of this lesson, students will be able to evaluate a technology using benefits, limitations and stakeholder needs.	**H:** Return to the automated production case and ask whether faster production automatically makes a change desirable for everyone. List the people whose experience might differ. **ID:** Explain evaluation as judging against criteria using evidence. Introduce effects on workers, manufacturers, customers and the environment. Discuss initial investment, training, accuracy, waste and repair, distinguishing a supported benefit from a promotional claim. **WD:** Use a fictional school deciding whether to outsource a small batch of display parts. Compare two processes against quantity, lead time, material waste and cost using supplied case data. Model a recommendation that acknowledges a	**Support:** Provide a partly completed grid and explicit stakeholder cards. Rehearse a recommendation orally using evidence, judgement and limitation prompts before students write independently. **Extension:** Students identify the most influential criterion, explain a sensitivity to quantity or budget and describe the additional evidence needed for a stronger decision.

			drawback and states an assumption. **YD:** Students use a decision grid to compare two manufacturing options for a defined brief. They identify two benefits, two limitations and the people affected, then recommend an option in a short paragraph. Partners check that the recommendation uses the supplied evidence rather than a general claim that newer is always better. **C:** Ask which change in the brief could reverse the recommendation. Check that students weigh competing criteria and recognise uncertainty in the case data.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.1 The impact of new and emerging technologies; 1.2 Critical evaluation
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?